

```

#include <stdio.h>

int main() {
    FILE *mcode = fopen("machinecode.txt", "w");
    char tac[20][20], quad[20][40], op, oper[10][10];
    int n;

    printf("Enter the number of lines: ");
    scanf("%d", &n);

    printf("Enter the code: ");
    for (int i = 0; i < n; i++) {
        scanf("%s", tac[i]);
        quad[i][1] = tac[i][2];

        if (tac[i][3] != '\\0') {
            quad[i][2] = tac[i][4];
            quad[i][0] = tac[i][3];
        } else {
            quad[i][2] = '=';
            quad[i][0] = '=';
        }
        quad[i][3] = tac[i][0];
    }

    printf("\nQuadruples: \n");
    printf("op\targ1\targ2\tresult\n");
    for (int i = 0; i < n; i++) {
        printf("\n");
        for (int j = 0; j < 4; j++) {
            printf("%c \t", quad[i][j]);
        }
    }
    for (int i = 0; i < n; i++) {
        fprintf(mcode, "MOV R%d, %c\n", i, quad[i][1]);
        op = quad[i][0];

        if (op != '=') {
            switch (op) {
                case '+':
                    fprintf(mcode, "ADD R%d, %c\n", i, quad[i][2]);
                    break;

                case '-':
                    fprintf(mcode, "SUB R%d, %c\n", i, quad[i][2]);
                    break;
                case '/':
                    fprintf(mcode, "DIV R%d, %c\n", i, quad[i][2]);
                    break;
                case '*':
                    fprintf(mcode, "MUL R%d, %c\n", i, quad[i][2]);
                    break;
            }
        }
        fprintf(mcode, "MOV %c, R%d\n", quad[i][3], i);
    }

    printf("\nTAC\n");
    for (int i = 0; i < n; i++) {
        printf("%s\n", tac[i]);
    }

    return 0;
}

```